

Title: Technical Brief: Jarvis Multimodal AI

Author: Pranava, Modern Public School

Institution: Rishabh Vihar, New Delhi

1. Objective: To develop a high-speed, local-first multimodal assistant that eliminates cloud latency and ensures data privacy.

2. Architecture: * Speech Recognition: OpenAI Whisper-base for real-time STT (Speech-to-Text).

- . Reasoning Engine: Microsoft Phi-3-mini (3.8B Parameters) for local LLM inference.**
- . Optimization: 4-bit GGUF quantization to achieve sub-1.5s response time.**

3. Key Performance: * Latency: Average response time of 1.2 seconds on consumer-grade hardware.

- . Privacy: 100% offline execution; zero external data leakage.**

4. Conclusion: Jarvis demonstrates that Small Language Models (SLMs) can deliver executive-level assistant capabilities with minimal compute requirements.